# Distribution Board



Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High**

1 labs

**Covers these lab items:** Distribution Box (MCB, RCCB, Isolator)

## What it is used for

A distribution board distributes electrical power to outgoing circuits and provides switching and protective devices such as MCBs, RCCBs, and isolators. Correct inspection, circuit identification, and protection testing help maintain safe distribution and reduce electric-shock, overload, and fire risks.

## Main parts

- Enclosure and protective cover
- Main isolator or incoming switch
- Miniature circuit breakers (MCBs)
- Residual current circuit breaker (RCCB), where fitted
- Neutral and earth bars
- Busbars, terminals, labels, and cable entries

## Before you start

1. Only a qualified and authorised electrician should inspect, install, or test the distribution board.
2. Identify the supply source, nominal voltage, circuit schedule, and board rating from the approved drawings and labels.
3. Check the enclosure, cover, circuit labels, cable entries, and visible wiring for damage, moisture, overheating, or corrosion without touching exposed conductors.
4. Confirm that the work area is dry, adequately lit, and clear of obstructions.
5. For internal inspection or maintenance, isolate all relevant power sources, apply lockout/tagout where applicable, and verify absence of voltage with an appropriate tested instrument.
6. Use the required electrical PPE, insulated tools, and test equipment rated for the installation.

## How to use

1. Review the approved circuit schedule and identify the incoming supply and outgoing circuits before any work.
2. Confirm that the distribution board is suitable for the installation and that protective devices match the approved design.
3. With power safely isolated and absence of voltage verified, inspect terminals, busbars, neutral and earth connections for signs of looseness, overheating, or damage.
4. Check that cable sizes, terminations, protective-device ratings, and circuit labels agree with the approved design.
5. Verify that neutral and protective earth conductors are connected to the correct designated bars and are not interchanged.

6. Check enclosure bonding, cable glands, cover fit, and barriers to prevent accidental contact with live parts.
7. Carry out continuity, insulation resistance, polarity, and earthing checks as required by the approved test procedure and applicable electrical rules.
8. Test MCB and RCCB operation using the appropriate test method and instrument; use the RCCB test button only as an additional routine check where specified.
9. Reinstall all covers and barriers and confirm that no tools or loose materials remain inside the enclosure.
10. After inspection is complete and the area is clear, restore supply using the authorised switching procedure.
11. Verify circuit operation and check for abnormal tripping, heating, smell, or noise without opening an energised enclosure.
12. Record test results, identified defects, corrective actions, and any circuits that remain isolated.

## ⚠ Safety

- Never touch exposed conductors or work inside an energised distribution board unless a formally authorised procedure specifically permits the task and suitable controls are in place.
- Isolate all relevant sources, including backup supplies or generators, and verify absence of voltage before internal work.
- Do not rely solely on an isolator position or indicator lamp to confirm that a circuit is dead.
- Never bypass, oversize, or replace protective devices with incorrect ratings.
- Keep the enclosure closed and secured against access by unauthorised persons.
- Do not allow water, metal objects, or loose debris to enter the board.
- Follow applicable electrical codes, lockout/tagout procedures, and site emergency arrangements.

## Care & storage

- Keep the enclosure dry, clean, closed, and clearly labelled.
- Arrange periodic inspection and testing by qualified personnel according to applicable requirements.
- Investigate repeated breaker trips, RCCB operation, burning smells, discoloration, or abnormal heating promptly.
- Maintain an up-to-date circuit schedule and records of test results and modifications.
- Replace damaged breakers, terminals, covers, or wiring only with compatible approved components and after safe isolation.

## Common mistakes to avoid

- Working on the board without isolating and verifying absence of voltage.
- Using protective devices with incorrect ratings or bypassing them to stop tripping.
- Mixing neutral and earth conductors or leaving terminals loose.
- Restoring power before covers, barriers, and circuit labels are properly reinstated.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the*

*manufacturer's original manual / file.*